Example 2.30

Evaluate $\int \frac{x}{x^2+1}dx$

Solution:

Take $f(x) = x^2+1$

$\therefore \quad f'(x) = 2x$

$$\int \frac{x}{x^2+1}dx = \frac{1}{2}\int \frac{2x}{x^2+1}dx$$

$$= \frac{1}{2}\int \frac{f'(x)}{f(x)}dx$$

$$= \frac{1}{2}\log[f(x)]+c$$

$$= \frac{1}{2}\log|x^2+1|+c$$



$$\int \frac{nax^{n-1}}{ax^n+b}dx = \int \frac{d(ax^n+b)}{ax^n+b}$$

$$= \log|ax^n+b|+c$$

Example 2.31

Evaluate $\int \frac{x}{\sqrt{x^2+1}}dx$

Solution:

Take $f(x) = x^2+1$

$\therefore \quad f'(x) = 2x$

$$\int \frac{x}{\sqrt{x^2+1}}dx = \frac{1}{2}\int \frac{2x}{\sqrt{x^2+1}}dx$$

$$= \frac{1}{2}\int \frac{f'(x)}{\sqrt{f(x)}}dx$$

$$= \frac{1}{2}\left[2\sqrt{f(x)}\right]+c$$

$$= \sqrt{x^2+1}+c$$



$$\int \frac{nax^{n-1}}{\sqrt{ax^n+b}}dx = \int \frac{d(ax^n+b)}{\sqrt{ax^n+b}}$$

$$= 2\sqrt{ax^n+b}+c$$

Example 2.32

Evaluate $\int x\sqrt{x^2+1}\,dx$

Solution:

Take $f(x) = x^2+1$

$\therefore \quad f'(x) = 2x$

$$\int x\sqrt{x^2+1}\,dx = \frac{1}{2}\int (x^2+1)^{\frac{1}{2}}(2x)dx$$

$$= \frac{1}{2}\int [f(x)]^{\frac{1}{2}} f'(x)dx$$

$$= \frac{1}{2}\frac{[f(x)]^{\frac{3}{2}}}{\frac{3}{2}}+c$$

$$= \frac{1}{3}(x^2+1)^{\frac{3}{2}}+c$$

Example 2.33

Evaluate $\int \frac{x^3}{(x^2+1)^3}dx$

Solution:

Take $z = x^2+1$

$\therefore x^2 = z-1$ *and*

$dz = 2x\,dx$

$\Rightarrow \frac{dz}{2} = x\,dx$

$$\int \frac{x^3}{(x^2+1)^3}dx = \int \frac{x^2}{(x^2+1)^3}x\,dx$$

$$= \frac{1}{2}\int \frac{z-1}{z^3}dz$$

XII Std - Business Maths & Stat EM Chapter 2.indd 36 2/28/2020 5:38:50 PM